

Chapter 51. Humidification

Jean-Damien Ricard; Didier Dreyfuss

Humidification: Introduction

Humidification issues are overlooked by many clinicians in the intensive care unit (ICU). Because the need to heat and humidify inspired gases during mechanical ventilation is unanimously accepted, this process is considered the basic, supportive standard of care, about which there is no real debate. Yet, considerable controversy has surrounded central issues concerning humidification such as the level of adequate humidification and how to provide it, the influence of humidification devices on the incidence of ventilator-associated pneumonia, and certain patients and clinical situations and their requirements, such as the need for humidification during noninvasive ventilation. This may account for important differences in the practice of humidification between countries.^{1,2} Fortunately, renewed interest has emerged over the past decade, as indicated by several clinical studies that have helped settle some controversies. This chapter will review the reasons for conditioning inspired gases by recalling the normal process of heating and humidifying air during spontaneous breathing, the physical principles of humidification, and the consequences of inappropriate conditioning. Devices to achieve this conditioning are covered and their advantages and potential drawbacks discussed. Finally, practical guidelines are provided.

Rationale

As mentioned in the next section, the upper respiratory tract is responsible for most of the conditioning of the inspired gases. Important features of this conditioning (reviewed by Irlbeck³) include heat, humidification, and filtration, in order to deliver to the lower respiratory tract a warm (32°C [89.6°F]), humid (95% relative humidity), and pathogen-free and particle-free gas. The last step is achieved in the lower respiratory tract. During invasive mechanical ventilation, the endotracheal tube bypasses the upper respiratory tract. This places the burden of supplying heat and humidity to the cold and dry medical gases on the lower respiratory tract, a task for which it is poorly suited.⁴

Anatomy and Physiology of the Upper Airway

The upper respiratory tract (nose, mouth, nasopharynx, oropharynx, laryngopharynx, and larynx, but mainly the nose) is responsible for most of the conditioning of inspired gas. Anatomic structure and physiologic function of the nose are intimately linked. The highly vascular mucosa of the nose is ciliated and rich in mucosal glands and goblet cells. Three curved bony plates on the lateral side of each nasal cavity (the superior, middle, and inferior concha or turbinate bones), covered with a mucous membrane, ensure the important function of satisfactorily conditioning inspired gases. Their large surface area and position in relationship with the air current enable sufficient contact with the inspired gas. This mucous membrane also lines the paranasal sinuses, trachea, and bronchi but not the pharynx, which does not take part in the air-conditioning process.

Physical Principles of Humidification

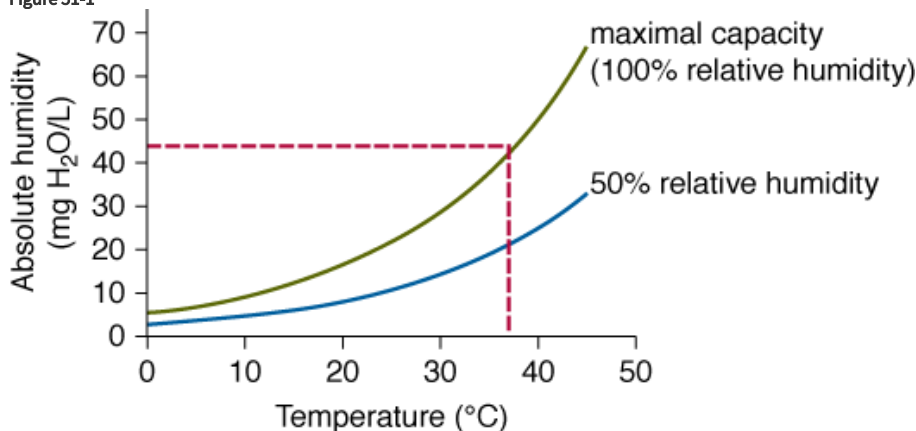
Humidity can be defined as the moisture content of the atmosphere and by extension, water present as vapor in a gas mixture. Vaporization indicates the change of a liquid (or a solid) to a gas or vapor. There is no strict difference between the terms *gas*

and *vapor*, although gas is generally used to describe a substance that appears in the gaseous state under standard conditions of pressure and temperature, and vapor to describe the gaseous state of a substance that appears ordinarily as a liquid or solid. Such a change of state requires a certain amount of energy or heat to overcome the van der Waals forces that bind the molecules together in the liquid state. They then escape the liquid as individual molecules of vapor. This amount of heat, specific for each substance, is known as the latent heat of vaporization of the substance.

Importantly, liquids can change to gases at temperatures below their boiling points. Vaporization of a liquid below its boiling point is called evaporation, and can occur at any temperature when the surface of a liquid is exposed in an unconfined space. Molecules of liquid that leave the surface of the liquid (because of sufficient kinetic energy) turn to vapor. They do so with their latent heat; the loss of heat causes the liquid's temperature to decrease. These molecules of vapor exert a certain amount of pressure known as vapor pressure. Vapor pressure is substance and temperature specific (i.e., each substance has a specific vapor pressure for each given temperature). At its boiling point, the vapor pressure of a liquid is equal to the atmospheric pressure. Therefore, vapor pressure of water at 100°C (212°F) is 760 mm Hg.

Measurements of humidity include absolute humidity and relative humidity. Absolute humidity is the mass of water vapor per unit volume of the gas mixture and is measured in milligrams of water per liter of gas (mg H₂O/L). Importantly, at any given temperature, a gas may contain only up to a certain amount of water vapor, corresponding to its maximal capacity at that temperature. Any extra water vapor will condense back into a liquid state. Relative humidity indicates the ratio of the actual water vapor content of the gas mixture to its total capacity at the given temperature. The amount of water vapor a gas can contain is directly proportional to the temperature of the gas (Fig. 51-1). For example, air with 50% relative humidity contains 22 mg of H₂O at 37°C (98.6°F), but only 8.5 mg of H₂O at 20°C (68°F).

Figure 51-1



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Absolute humidity (mass of water vapor) of air as a function of temperature at two relative humidities (100% and 50%). The *dotted lines* indicate the value of absolute humidity (44 mg H₂O/L) at core body temperature (37°C [98.6°F]) when air is fully saturated (100% relative humidity).

Physiologic Principles of Humidification

Physiologic humidification delivers inspired air at 37°C (98.6°F) and saturated with water vapor (100% relative humidity) to the lungs. Figure 51-1 indicates that in these conditions, the gas mixture contains 44 mg of H₂O per liter. The principle of humidification lies in the heat and humidity gradient that exists between the inspired gas and the respiratory tract.

As indicated above, humidification occurs first and mainly in the nose, then in the trachea and up to the second-generation bronchi. The large surface area of the turbinates, in combination with their irregular shape and protrusion into the nasal cavity

that create a turbulent flow, enables a rapid transfer of heat from the mucosa to the inspired air. This heat gain is sufficient to allow water to evaporate from the mucosa into the inspired gas, enabling its humidification. As a result, inspired gas is warmed and humidified and nasal mucosal temperature decreases from 37°C (98.6°F) to 31°C (87.8°F).

During expiration, the opposite occurs. The gas leaves the alveolus at 37°C (98.6°F) with 44 mg/L of H₂O, and transfers heat to the cooler upper airway. As its temperature decreases, so does its maximal water content capacity. Condensation on the mucosa then occurs. At the end of expiration, approximately one-third of the heat and humidity has been transferred back to the mucosa.

Until the inspired gas has reached 37°C (98.6°F) with 44 mg/L water content, heat and water are lost from the respiratory tract. The point at which inspired gas reaches core temperature and 100% relative humidity is called the isothermic saturation boundary. It is usually situated in the second-generation bronchi. After this point, temperature and humidity of inspired gas remain constant, to ensure adequate gas exchange in the alveolus. Depending on climatic conditions, the isothermic saturation boundary may move up or down the respiratory tract.

Pathophysiology of the Upper Airway

Underhumidification

Underhumidification results from the bypassing of the upper airway that follows tracheal intubation and from the absence or inadequacy of conditioning of the dry, compressed, medical gas. Delivery to the lower respiratory tract of insufficiently warmed and humidified gas displaces the isothermic saturation boundary further down the bronchial tree. Moisture and heat loss ensues, inducing structural damages of the respiratory tract that may lead to physiologic and functional impairments with clinical repercussions.

Effects of Moisture Loss

Moisture loss from the respiratory tract is, under normal conditions, a physiologic phenomenon (approximately 150 mL water) and contributes to the body's insensible water loss. The loss may be much greater if minute ventilation is increased, or if the gradient between the water content of inspired and expired air is large or both. This situation is encountered during mechanical ventilation and was first described during general anesthesia. In one of the first comprehensive studies on this topic, Burton described the reduction of mucociliary function after several hours of dry gas ventilation. When heat and humidity were added, mucociliary function was restored.⁵ Numerous studies have been performed subsequently and exhaustively reviewed.⁴ The adverse effects of inadequate humidification comprise structural damages such as loss of ciliary function, cilia destruction, damage to mucus glands, cytoplasmic and nuclear degeneration, cellular desquamation, mucosal ulceration, changes in tracheal cytology, and loss of surfactant. Such structural damage has functional repercussions: destruction of the mucociliary escalator, increase in sputum viscosity, increased airway resistance, decreased pulmonary compliance, reduced functional capacity, and increased pulmonary shunting. Clinical consequences include retained secretions, mucus plugging, atelectasis, increased work of breathing, hypothermia, and hypoxemia.³ In clinical practice, the worst and most feared adverse effect of underhumidification is life-threatening endotracheal tube occlusion, which requires urgent endotracheal tube replacement (see section on [Effectiveness and Outcome](#)).

Effect of Heat Loss

During general anesthesia, the amount of heat loss through the respiratory tract is very small in comparison with that of other sources (e.g., the operative field and skin). Two mechanisms may account for this loss. Heat is transferred from the respiratory mucosa to the inspired gases, and the amount depends on the patient's minute ventilation, the temperature gradient between the respiratory tract and the inspired gas, and the latter's specific heat. Given the very low specific heat of air, minimal heat is

required to raise its temperature. The second and major mechanism by which heat is lost is vaporization (change of liquid water of the respiratory mucosa into vapor and its transfer to the inspired gas). Given the high specific heat of water, a significant amount of heat is required to raise the relative humidity of inspired gases. The amount of heat is further increased when the relative humidity of the inspired gas is very low (i.e., dry medical gas). These considerations may be important during prolonged surgery in infants and young children, keeping in mind that skin, operative site, and fluid administration are the major contributors to the heat loss.

Overhumidification

Because heat and moisture exchangers (HMEs) act passively (see below), they cannot deliver excessive heat or water to the respiratory tract. Consequently, the adverse effects described below are only possible with heated humidifiers and aerosol humidifiers and are now seldom encountered.⁶

Excess Water

The volume of water that can be added to the airway mucosa depends on the state it is in. With heating and humidifying devices, water is in a molecular form (vapor) and is therefore unlikely to result in overhumidification, given the amount of water vapor contained in inspired gases. To significantly increase this amount would require excessive heating of the inspired gas (way above body temperature), thus causing thermal injury before overhydration. Effects of excess water have been reported in the past with the use of aerosol humidifiers,⁷ but clinicians are now well aware of these side effects.

Excess Heat

Reports of lung injury secondary to excessive heat are rare⁶ and are related to the misuse of heated humidifiers.⁸

Methods of Humidification

Although many different types of humidifiers have been developed in the past, clinicians mainly use two types of devices to condition inspired gas during mechanical ventilation: heated humidifiers and HMEs.

Heated Humidifiers

The general working principle of heated humidifiers is to heat water contained in a humidification chamber that humidifies inspired gas passing through it (Fig. 51-2).⁹ Heating the water enables evaporation to occur, and the gas leaves the chamber saturated with water vapor. Evaporation depends on the surface area over which it occurs, the temperature of the liquid water, and the magnitude of the vapor pressure above the water surface. It can therefore be increased by augmenting the contact of inspired gas and water (larger humidification chamber), by raising the temperature of the liquid water, and by increasing the mass flow of gas above the water surface so as to decrease the vapor pressure. Once the inspired gas has left the humidifier, it cools along the breathing circuit before reaching the patient. Thus, the temperature one wished to achieve at the patient's level will not be reached. There are two ways to help overcome this problem. Inspired gas can be heated in the chamber to above the temperature one wishes to deliver to a patient, to take into account the heat loss across the circuit. Abundant condensation, however, will occur inside the tubing, providing an ideal reservoir for bacterial colonization^{10,11} and will need to be drained, exposing health care workers to contaminated fluids and placing other patients at risk of cross-contamination.¹² To prevent this condensation (that results from the decrease in the temperature of the gas whose maximum vapor-carrying capacity is therefore reduced), breathing circuits can be heated by placing electric heater wires either in the wall or in the lumen of the circuit.



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Heated humidifier with a heated breathing circuit. Two temperature probes monitor gas temperature at the exit of the humidifier chamber and at the Y-piece before the endotracheal tube (*black arrows*).

New devices are now servocontrolled by means of temperature sensors placed at the exit of the humidification chamber and just before the catheter mount (or flex tube). Because only the temperature is monitored (and not the humidity), however, these devices may become faulty in certain circumstances (see below).^{13,14}

Heat and Moisture Exchangers

HMEs have undergone considerable development since their first description in the mid-1950s.¹⁵ The basic principle of all HMEs lies in their capacity to retain heat and moisture during expiration and deliver these to the incoming dry medical gas during subsequent inspiration (*Fig. 51-3*).¹⁶ This passive function has been achieved by different mechanisms. Metal elements of the first condensers were replaced by disposable foam, plastic, or paper. Increased moisture output was further achieved by coating the condenser element with a hygroscopic chemical (calcium or lithium chloride), which chemically adsorbs expired water vapor that is then returned to the inspired gas. These HMEs were called hygroscopic condensers. Derived from the field of filtration, hydrophobic HMEs were also used to heat and humidify inspired gas. The very large surface area of pleated, water-repellent ceramic of the first hydrophobic HMEs along with its low thermal conductivity enabled an important temperature gradient to develop within the HME that favored heat and moisture retention. Finally hygroscopic and hydrophobic elements were used in a single HME to create a “combined” HME. Hygrometric performance of these various HMEs differ considerably,¹⁷ with hydrophobic HMEs exhibiting the lowest humidity outputs (with the attendant risk of endotracheal tube occlusion) and combined HMEs the highest, although recent measurements indicate that some purely hygroscopic HMEs provide levels of

absolute humidity comparable to those of combined HMEs.^{18,19} New devices have been designed to boost the humidifying performance of HMEs,²⁰ although their actual clinical benefit is uncertain.

Figure 51-3



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A combined (hydrophobic and hygroscopic) heat and moisture exchanger. *Dark arrows* indicate the presence of dripping wet condensation in the flexible tubing (the bright reflection seen on the wall of the tubing) suggesting that adequate humidity is being delivered by the heat and moisture exchanger (see text for details). Note that the heat and moisture exchanger is positioned vertically above the endotracheal tube, thus limiting the amount of secretions refluxing from the tube to the heat and moisture exchanger.

Application

Based on the above principles, supplying heat and humidity to inspired gases is mandatory when the upper airway is bypassed during tracheal intubation. This supply must be provided as soon as the patient is connected to the ventilator, either by an HME or by a heated humidifier. Apart from certain specific situations (see “[Practical Steps](#)”), both HMEs and heated humidifiers serve equally well. Despite this, practices differ considerably among countries.^{1,21} Rationalizing the choice of a given humidifying device should help decrease costs in mechanical ventilation without impeding quality of care.

Effectiveness and Outcome

Several important aspects must be taken into account when analyzing the effectiveness and outcomes of inspired gas conditioning, including:

The avoidance of endotracheal tube occlusion (the worst and most feared complication of inadequate gas conditioning). This aspect is directly linked to the performance of the humidifying device in terms of heat and humidity delivery;

The avoidance of spreading microorganisms (especially multidrug-resistant organisms). This is directly linked to the humidifying device's capacity to prevent contamination of the respiratory tubing (Fig. 51-4 and Fig. 51-5);

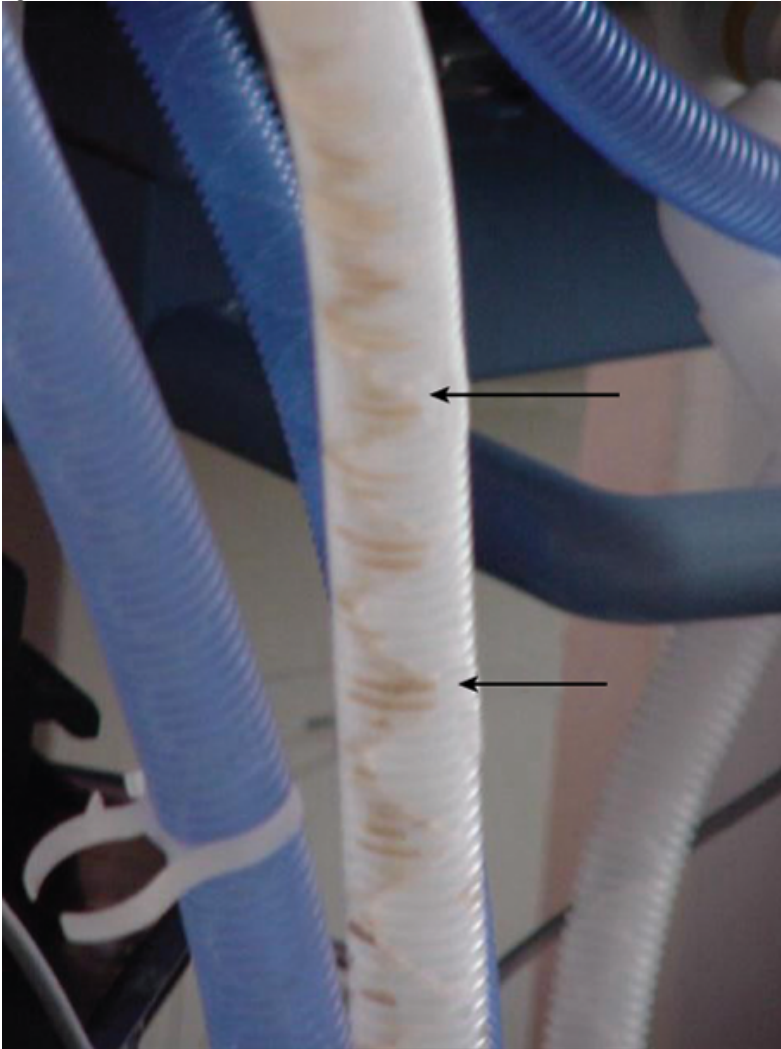
The addition of minimal resistance and dead space;

The practicability of use of the device;

The minimal maintenance necessary to ensure optimal use of the device; and

The minimal cost associated with the purchase and the long-term use of the device.

Figure 51-4



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A heavily contaminated breathing circuit.

Figure 51-5



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A dual heated respiratory circuit. The new generation expiratory limb (*white tubing*) allows water vapor to diffuse through the tubing wall, thus minimizing condensate formation in the limb.

Using this list, one can define the characteristics of the ideal humidifier: a device that provides adequate levels of humidification whatever the ventilator and patient conditions; one that does so automatically; one that is safe to operate (i.e., is electric-shock free, with no or a limited number of connections that can become faulty); one that protects the environment from the patient's pathogens; one that is easy to use; and one that requires low maintenance and is inexpensive.

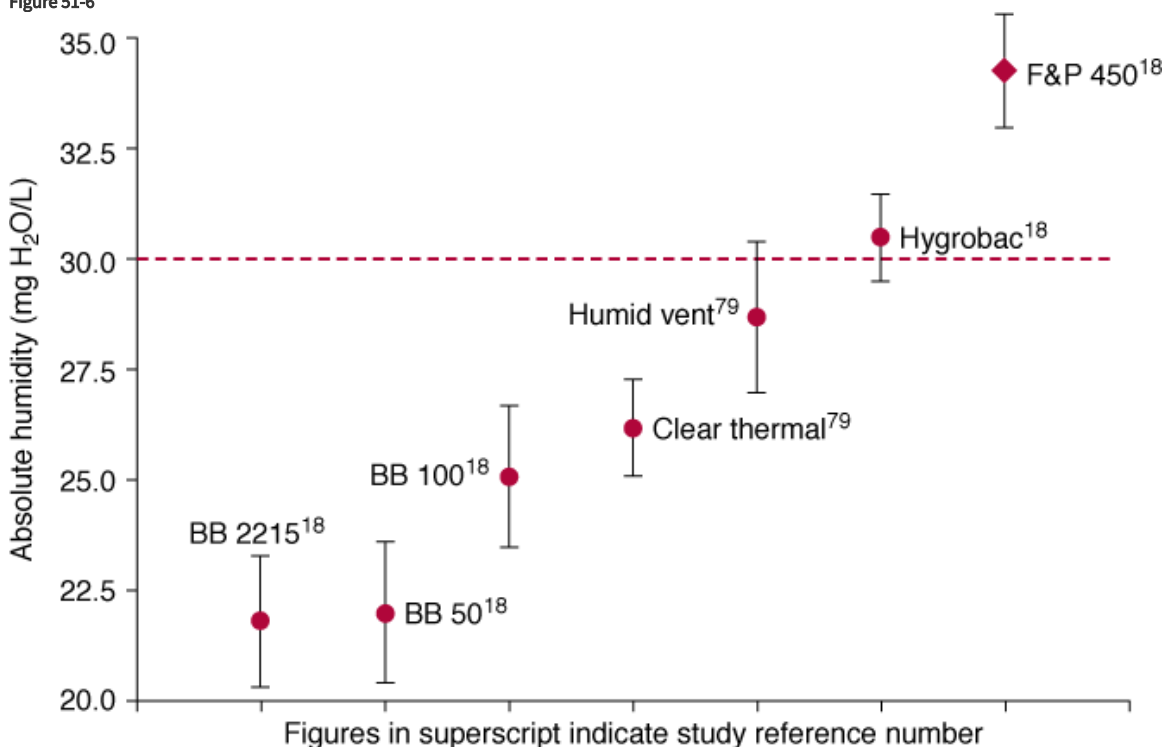
Ensuring Adequate Humidification

The adequate level of humidification can be defined as a level at which there is no excessive heat or water loss by the respiratory tract. The difficulty arises when one tries to set the minimum value of absolute humidity a device should deliver. This may be useful when selecting and comparing different devices. Although ranges as wide as 25 to 35 mg H₂O/L have been suggested in the past,²² the figure of 30 mg H₂O/L is recommended today.²⁴ That is, a clinician wishing to select an appropriate humidifying device should make sure that the device delivers at least 30 mg H₂O/L of absolute humidity. A recent study indicates that this output is not achieved by all available HMEs.²³

In the past, HMEs and heated humidifiers were not equivalent in terms of humidity output. Back in the late 1980s, several studies alerted clinicians to the risk of endotracheal tube occlusions associated with the use of HMEs.^{24–27} These studies, however, all used purely hydrophobic HMEs, whose measured performance displayed low values of absolute humidity.^{17,18,28,29}

Although heated humidifiers indisputably deliver greater values of absolute humidity than HMEs^{17,28,29} (Fig. 51-6), there is to date no evidence of any benefit of these greater values in terms of clinical outcome, including endotracheal tube occlusion. Table 51-1 displays the incidence of endotracheal tube occlusion reported in recently published studies. It is important to note that: (a) endotracheal tube occlusion rates have drastically dropped in comparison with earlier studies;^{25–27} (b) endotracheal tube occlusion is also reported with the use of heated humidifiers; and (c) the incidence of endotracheal tube occlusion no longer appears to be greater with HMEs than with heated humidifiers. In one study, the incidence was greater with heated humidifiers than with HMEs.³⁰

Figure 51-6



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Values of absolute humidity delivered by several heat and moisture exchangers and one heated humidifier. The *dotted horizontal line* delineates the minimum value of humidity (30 mg H₂O/L) a device should deliver. Note the discrepancies that exist among heat and moisture exchangers (*red circles*). The HMEs exhibiting the lowest values of absolute humidity are those most strongly associated with endotracheal tube occlusions. The heated humidifier (*red diamond*) exhibited the highest value for absolute humidity.

Table 51-1: Endotracheal Tube Occlusion Rates in Studies Comparing Heated Humidifier and Heat and Moisture Exchangers

		No. of ETT Obstructions	
Reference	No. of Patients	HH	HME
Branson et al 1993 ⁷⁵	120	0	0
Dreyfuss et al 1995 ⁵⁴	131	0	1
Branson et al 1996 ⁷⁶	200	0	0
Villafane et al 1996 ³¹	23	1	0
Boots et al 1997 ⁵⁷	116	0	0
Hurni et al 1997 ⁷⁸	115	1	0
Kollef et al 1998 ⁵⁸	310	0	0
Thomachot et al 1998 ⁷⁴	29	0	0
Kirton et al 1998 ⁵⁶	280	1	0
Lacherade et al 2002 ³²	370	5	1
Jaber et al 2004 ⁹³	60	2	1

Abbreviations: ETT, endotracheal tube; HH, heated humidifier; HME, heat and moisture exchanger.

The reasons for this unusually high rate of endotracheal occlusions deserve particular attention. As indicated above (see “[Methods of Humidification](#)”), new heated humidifiers are servocontrolled. Because the temperature of the incoming gas mixture has a direct effect on the heating mechanism of the plate that warms the humidification chamber, if the temperature is too high, the plate no longer heats, resulting in very low levels of absolute humidity,^{13,14} sometimes below 20 mg H₂O/L.^{13,14} These very low values, well below the 44 mg H₂O/L expected with these devices, are encountered when the temperature of the gas exiting the ventilator is high and/or when the ambient temperature is high.^{13,14} It is notable that depending on the type of ICU ventilator, the output ventilator temperature may vary from 26°C to 35°C (78.8°F to 95°F).^{13,14} An automatic compensation system has been designed that seems to be effective in overcoming this worrisome problem.^{14,31} This system is available on some, but not all, heated humidifiers. If not, HMEs should be used.

Resistance

Both types of humidifying devices (HMEs and heated humidifiers) marginally increase resistance to flow in the respiratory circuit^{32–35} with no impact on auto-positive end-expiratory pressure, even in patients with chronic obstructive pulmonary disease (COPD).³⁶ Thus, resistance to flow of humidifying devices during invasive mechanical ventilation is not a major problem clinically.

Dead Space

Because of their internal volume, HMEs increase dead space in the ventilator circuit.^{35,37,38} This increase is directly proportional to the internal volume of the HME, which may vary between 30 mL and 95 mL.³⁸ The impact of dead space on work of breathing and arterial CO₂ depends on the mode of ventilation and tidal volume used. During volume-assist control ventilation, internal dead space will only influence PaCO₂ and pH: the smaller the tidal volume used (in combination with an HME with a large internal volume), the greater the impact on arterial CO₂ and pH.^{39,40} During invasive and noninvasive pressure-support ventilation, use of HMEs may increase work of breathing in COPD and in difficult-to-wean patients.^{34,41–45} This increase in work of breathing, however, is easily overcome by increasing the level of pressure support by 5 to 10 cm H₂O.^{34,41,43,44}

Microbiologic Outcome

Nosocomial Pneumonia

The problem of nosocomial infection related to respiratory equipment is not new, and older inhalation therapy equipment posed a considerable challenge in ICUs in the past.^{46–49} Similarly, it has been suggested that contamination of respiratory tubing used with heated humidifiers may be a risk factor for ventilator-associated pneumonia.^{11,50} This hypothesis stemmed from the rapid and considerable bacterial colonization of respiratory tubing encountered with use of heated humidifiers.^{10–12,51} Because HMEs prevent bacterial contamination of respiratory tubing,^{52,53} the question arose as to whether, or not, the incidence of ventilator-associated pneumonia was greater with heated humidifiers.⁵⁴ There is now overwhelming evidence that the method of humidification does not influence occurrence of ventilator-associated pneumonia.^{30,52,55–58} This is not very surprising considering the pathogenesis, mainly the silent aspiration of contaminated gastric and oropharyngeal secretions.⁵⁹ Consistent with this reasoning, the incidence of ventilator-associated pneumonia is not affected by the duration of use of HMEs, whether changed every 48 hours,⁶⁰ 72 hours,¹⁸ or only once a week.⁶¹ Similarly, the type of HME does not influence the rate of ventilator-associated pneumonia.^{62,63} Despite these facts, some have recently recommended they be preferred to heated humidifiers to prevent nosocomial pneumonia.⁶⁴

Cross-Contamination

Although use of heated humidifiers is not associated with a greater incidence of ventilator-associated pneumonia, these devices do have the potential for cross-infection.¹² Craven et al studied ventilator-circuit colonization during the first 24 hours after a circuit change. They found a rapid colonization: 33% of circuits were colonized at 2 hours, 64% at 12 hours, and 80% at 24 hours. The median level of colonization at 24 hours was 7×10^4 organisms/mL.¹¹ Bacteria isolated in respiratory tubing condensates correlate with the microorganisms found in patients' tracheobronchial secretions.^{11,51,55} These highly contaminated condensates should therefore be handled as infectious waste and regularly emptied.¹¹ Repeating these septic

maneuvers several times a day increases the risk of cross-infection, especially when patients are colonized with multidrug-resistant bacteria.⁶⁵

Preventing respiratory tubing contamination confers on HMEs a great advantage over heated humidifiers.^{19,52,55} Attempts have been made to reduce formation of condensation in the circuits by heating both the inspiratory and the expiratory limb of the circuit. Such equipment needs to be evaluated in the clinical setting. A preliminary report found that condensation still occurred in up to 50% of patients ventilated with these heated circuits,⁶⁵ leading to substantial bacterial colonization (see Fig. 51-4). A modified expiratory limb enabling evaporation through the wall of the circuit considerably decreases occurrence of condensation in the circuit (see Fig. 51-5).

Practicability and Maintenance

For obvious reasons, HMEs are far easier to handle than heated humidifiers. They require no maintenance and several studies indicate that they reduce staff work load.^{52,56,66} Several technical improvements brought to the new heated humidifiers (servocontrol, automatic water-filling system, heated respiratory circuits, and so on) have considerably reduced the time required to check and operate them.

Safety

Clinicians are well aware of the hazards associated with HMEs (mainly endotracheal tube occlusion),¹ but may be less aware of the hazards associated with heated humidifiers. Some recently described hazards deserve particular attention. Burns to the patients and breathing circuit meltdowns have occurred in the past because of heated wires.⁶⁷⁻⁶⁹ More frequent and more recent is the risk of ventilator shutdown because of rain out from heated humidifiers.⁷⁰⁻⁷²

Costs

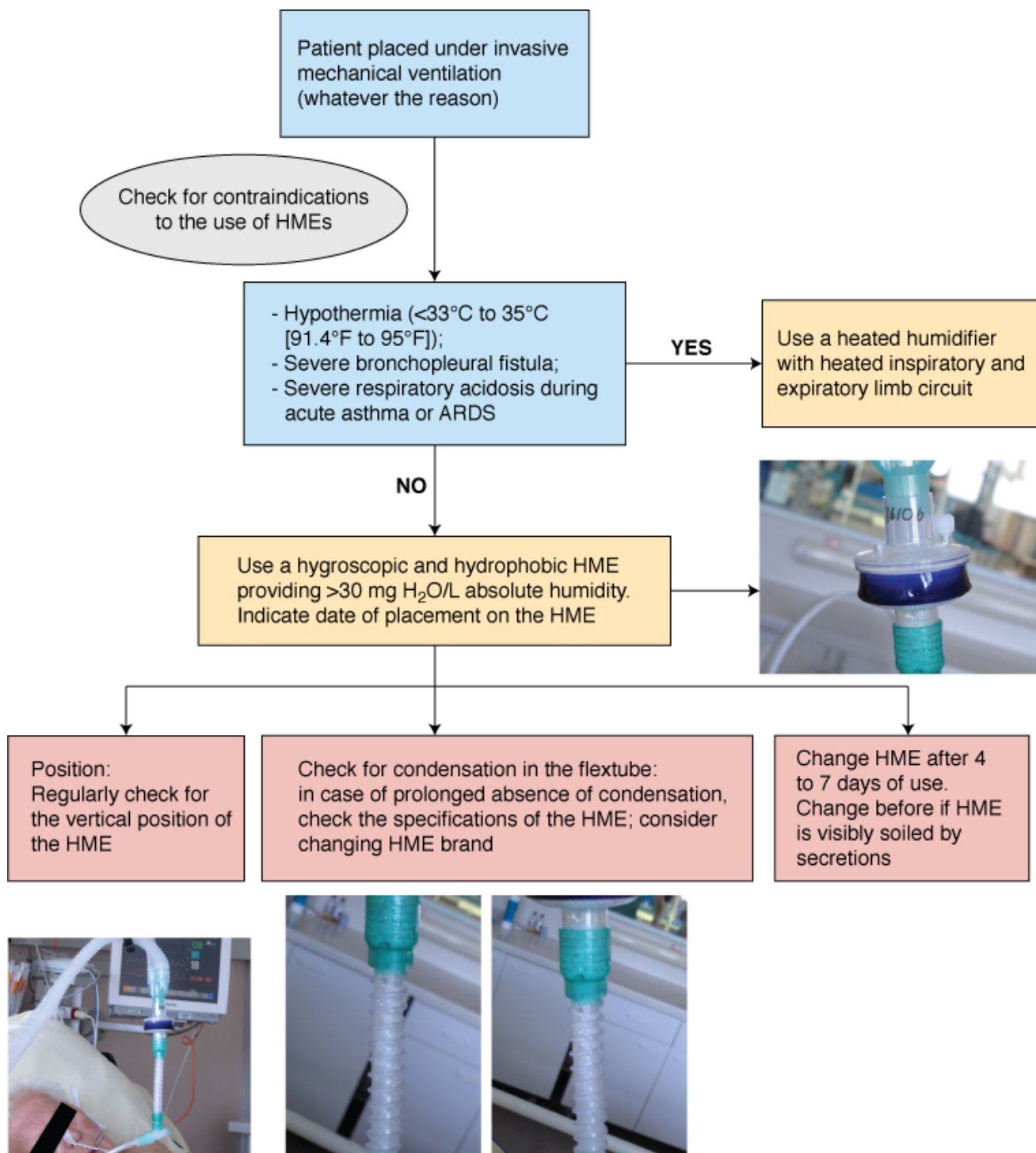
Several studies have compared the costs of humidification with HMEs with those of heated humidifiers.^{52,54-56,66,73,74} All conclude that considerable cost savings can be achieved by using HMEs, not only because heated humidifiers are expensive to buy, but also because they are expensive to run (especially because of the price of heated respiratory tubing). Costs can be further reduced by extending the use of HMEs; several studies indicate that these devices may be used without change for up to a week (see below).^{56,61,75}

Practical Steps

Invasive Mechanical Ventilation

An algorithm for heating and humidifying inspired gases during invasive mechanical ventilation is proposed in Figure 51-7.

Figure 51-7



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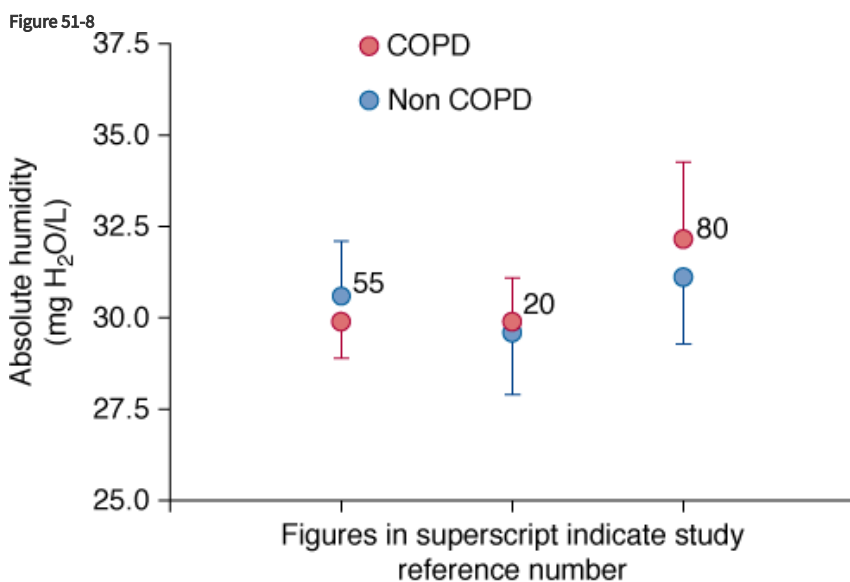
Algorithm for heating and humidifying inspired gases during invasive mechanical ventilation. *ARDS*, acute respiratory distress syndrome; *HME*, heat and moisture exchange.

Which Device for Which Patients

There are very few and rare contraindications to the use of HMEs. Because they act passively, the amount of heat and moisture they deliver to the inspired gases depends on the amount of heat and moisture they retain during expiration. Therefore, patients with profound hypothermia or important bronchopleural fistulas should preferably be ventilated with a heated humidifier. In addition, a heated humidifier may be preferable in patients ventilated for acute asthma or acute respiratory distress syndrome, in whom a drastic decrease in tidal volume induces extreme respiratory acidosis (to avoid increased dead

space with HMEs).^{39,40} A heated humidifier can be used until the patient's respiratory condition improves sufficiently to enable the use of an HME. Apart from these specific situations, HMEs should be considered first to heat and humidify inspired gases of all ventilated patients. They are as effective as heated humidifiers, and are much cheaper and much easier to use. Importantly, medical as well as surgical patients can benefit from them.

Despite the former practice of restricting their use to patients without a history of respiratory disease and only for the first 5 days of mechanical ventilation,^{74,76} several studies have clearly shown that they may be used in any ventilated patient, even in patients with COPD and for any length of mechanical ventilation. Because patients with COPD represent the most important subgroup of ventilated patients in the United States,⁷⁷ and because their duration of ventilation is longer than that of other patients,⁷⁷ adequate humidification is probably more critical in such patients than in patients ventilated for shorter periods. Numerous studies show that long-term mechanical ventilation can be safely conducted in such patients with HMEs.^{19,55,56,75,78–80} Three studies have specifically compared humidity output of HMEs in patients with and without COPD and consistently found that the measured values for absolute humidity were very similar (Fig. 51-8).^{19,75,80}



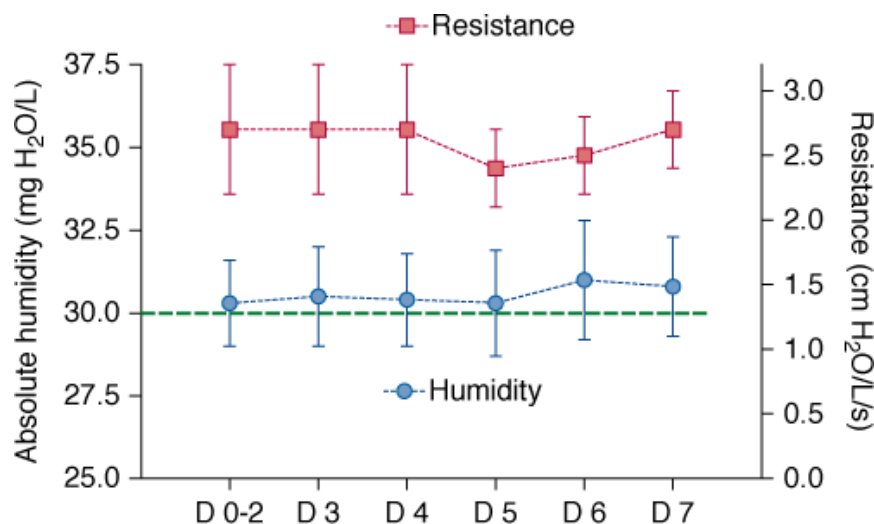
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Comparison of the absolute humidity delivered by heat and moisture exchangers in patients with and without chronic obstructive pulmonary disease (COPD). Values are similar, indicating that these devices are suitable for patients with COPD undergoing mechanical ventilation.

Frequency of Heat and Moisture Exchange Replacement

There is now considerable evidence that HMEs may be used for longer than the 24 hours recommended by manufacturers.^{18,19,56,60,61,73,75,79–82} Compelling results stem from rigorous clinical evaluation^{56,60} or extensive bedside measurement of humidity delivery.^{18,19,61,75,79–81} Figure 51-9 shows that humidity delivery of a combined HME (Hygrobac, DAR, Mirandola, Italy) used for 7 days without change was remarkably stable throughout this period. Progressive clogging of the device with tracheal secretions (that seriously increase resistance to airflow) could have been a drawback. Repeated measurements of the resistance of the HME over 7 days indicate that this phenomenon did not occur (Fig. 51-9).⁷⁵

Figure 51-9



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Absolute humidity (*left Y axis*) and resistance (*right Y axis*) measured over time in combined heat and moisture exchangers used for 7 days without change. Note the perfect stability of the humidity output and the lack of increase in resistance over time.

(Adapted, with permission, from Ricard et al.⁵³)

Keeping the HME vertically above the tracheal tube (see Fig. 51-3) (and having nurses and doctors repeatedly check the position of the HME) prevents secretions from refluxing from the tracheal tube and obstructing the HME.^{19,75} Use of HMEs may be extended in both medical patients (including those with COPD^{19,75,79,80}) and surgical patients.^{18,61,73,82} This practice is now widely accepted,⁸³ and recommendations have been made to change HMEs only once a week.⁶⁴ In our own published⁷⁵ and unpublished experience,⁷⁵ we have been changing HMEs only once a week since 1997, and have not encountered endotracheal tube occlusions.

Noninvasive Mechanical Ventilation

The question of heat and humidification requirements during noninvasive ventilation has gone unanswered until very recently. The 2001 International Consensus Conference on Noninvasive Ventilation made no recommendation as to whether a humidifying device should be used or not.⁸⁴ This absence stems from the uncertainties regarding the level of humidification required during noninvasive ventilation (given the fact that the upper airways are not bypassed) and that can be achieved during noninvasive ventilation with HMEs and heated humidifiers.

This aspect has received attention in a recent study that compared the water content of gas delivered obtained during noninvasive ventilation with either an HME or a heated humidifier, and the comfort or discomfort experienced by healthy subjects.⁸⁵ In the absence of humidification, water content was very low when an ICU ventilator was used (5 mg H₂O/L), but equivalent to ambient air hygrometry with a turbine ventilator at minimal fractional inspired oxygen concentration (F_IO₂) (13 mg H₂O/L). HME and heated humidifier had comparable performances (25 to 30 mg H₂O/L) although HME's effectiveness was reduced with leaks (15 mg H₂O/L). During continuous positive airway pressure, dry gases (5 mg H₂O/L) were less tolerated than humidified gases. Gases humidified at 15 or 30 mg H₂O/L were equally tolerated. These data indicate that in favorable conditions, HME and heated humidifiers provide equivalent comfort to the patient through similar water content to the inspired gases.

This study contrasts with some physiologic studies of the influence of the humidifying device on respiratory variables. In a randomized crossover study in nine patients receiving noninvasive ventilation for moderate to severe acute hypercapnic

respiratory failure, a significant increase in work of breathing was found with HMEs in comparison with heated humidifiers), although this did not significantly impact the partial pressure of arterial carbon dioxide (PaCO_2).⁴⁴ The dead space of HMEs is obviously responsible for these observations and a recent study showed that use of an HME with a very small dead space does not alter respiratory variables, compared with heated humidifiers.⁸⁶

The question that arises is whether the choice of a given humidifying device during noninvasive ventilation affects outcome, especially the intubation rate. To address this issue, a large multicenter randomized trial tested the hypothesis that the use of HMEs would increase intubation rates.⁸⁷ During 13 months, 247 patients were randomized to noninvasive ventilation either with an HME or a heated humidifier. The intubation rate (failure of noninvasive ventilation) was higher with a heated humidifier than with an HME, although the difference did not reach statistical significance (37.6% vs. 30.6%; $p = 0.31$). Mortality tended to be higher in the heated humidifier group (21.5% vs. 14.1%; $p = 0.18$). PaCO_2 tended to be lower in the heated humidifier group (66 vs. 72 mm Hg; $p = 0.08$).⁸⁷ These preliminary results suggest that the observed increases in work of breathing, minute ventilation, and PaCO_2 in carefully conducted physiologic studies (but on a small number of selected patients) may only play a marginal role in the clinical setting. Taken together, it can now be stated that noninvasive ventilation can be safely and efficiently performed with HMEs, provided attention is given to minimize leaks and that a small internal volume HME is chosen.

Adjustments at the Bedside and Troubleshooting

Assessing humidification at the bedside is desirable, if not essential, for at least two reasons: (a) a life-threatening tracheal tube occlusion²⁴ may occur without any precursory clinical signs of insufficient humidification; and (b) devices may not deliver, in some instances, the heat and humidity that is expected of them, because they are either malfunctioning¹⁴ or performance in the clinical setting does not attain that stated by the manufacturer.^{14,80}

Such assessment raises two questions: (a) Is the device delivering enough heat and humidity (according to the standards)? (b) Is the amount of heat and moisture delivered appropriate for a given patient?

Assessing Heat and Moisture Exchangers and Heated Humidifiers at the Bedside

A simple means of evaluating the humidity delivered by a given device is to assess the amount of condensation seen in the flexible tubing that connects either the Y-piece (when using a heated humidifier) or the HME to the endotracheal tube.^{14,17,88} Indeed, the amount of condensation seen in the flexible tubing is positively correlated with the absolute humidity measured at the bedside.^{14,17} If the tubing remains constantly dry over time, or only very few droplets of water are seen, then absolute humidity delivered by a device is probably below 25 mg $\text{H}_2\text{O}/\text{L}$, and the patient is at risk of endotracheal tube occlusion. When numerous droplets are seen in the tubing or if it is dripping wet, the device is probably delivering sufficient absolute humidity to prevent endotracheal tube occlusion.¹⁷

Assessing Adequacy of Inspired Gas Conditioning

Although it is conceivable that some patients have special humidification needs, the prevailing literature indicates that heated humidifiers and HMEs equally meet humidification requirements in the vast majority of ventilated patients. Therefore, monitoring the characteristics of secretions (thick and tenacious, or watery and abundant) as a guide to insufficient or excessive humidification may be of limited value, because changes in mucus characteristics may be entirely caused by the patient's condition (respiratory status, fluid balance, and so on), and not a consequence of the humidifying device. It has been shown that air humidification with either an HME or a heated humidifier has similar effects on mucus rheologic properties, contact angle, and transportability by cilia in patients undergoing mechanical ventilation.⁸⁹

Important Unknowns

The preliminary results showing a trend toward higher intubation and mortality rates following noninvasive ventilation with a heated humidifier⁸⁷ strongly bring into question the use of these devices during noninvasive ventilation, at least in patients for whom PaCO_2 is not the main problem. A question unanswered until recently was whether a humidification device was necessary in every single patient undergoing noninvasive ventilation. The above-mentioned study by Lellouche et al⁸⁵ provides some important clues. If an ICU ventilator is used, then additional heat and moisture is essential to ensure subject comfort, and this can be achieved equally by an HME or a heated humidifier. When a turbine ventilator is used, the hygrometry obtained is close to that of ambient air, and one should ask whether inspired gases are too dry or not. Unfortunately, because this study was conducted in healthy subjects, it remains unknown if its results are applicable to the clinical setting. One can legitimately hypothesize that the level of humidification that provides comfort to a healthy subject differs from that required for a patient with COPD suffering from acute respiratory failure.

The Future

In the field of acute respiratory failure management, underhumidification is no longer the main threat associated with invasive or noninvasive mechanical ventilation. The challenge now facing clinicians is the appropriate management (in terms of humidification) of a much larger population of patients that breathe spontaneously but require additional oxygen. Despite the millions of patients receiving external oxygen every day around the world, little data are available on their needs in terms of humidification. If very low flows of oxygen do not seem to be associated with marked discomfort, recent data indicate that cold-water bubble humidifiers perform poorly during oxygen therapy.⁹⁰ Additional heat and moisture, provided by a heated humidifier, only partially relief the dryness of mouth provoked in patients by dry oxygen.⁹⁰ Little is known when higher flows are used with a face mask. A new technique, called *high-flow nasal cannula oxygen*, has arisen from the neonatal and pediatric experience with very high-flow oxygen therapy. It combines a heated humidifier and an air-oxygen blender allowing flows up to 60 L/min with 100% FI_{O_2} (Optiflow, Fisher & Paykel, Auckland, New Zealand). One must bear in mind that patients' inspiratory flow during respiratory distress may exceed 60 L/min.⁹¹ Thus, conventional oxygen therapy, even at its maximum flow (15 L/min), will cover only a small fraction of a patient's inspiratory flow in terms of oxygen and humidification. With a high-flow nasal cannula, better matching between an adequately heated and humidified oxygen flow and a patient's inspiratory demand is obtained. Promising results have been reported with this technique in ICU patients in acute respiratory failure,^{92,93} as well as in patients presenting to the emergency department with hypoxemic respiratory failure.⁹⁴

Summary and Conclusion

Adding heat and moisture to the inspired gas during invasive mechanical ventilation is mandatory. This can be efficiently achieved either by heated humidifiers or by HMEs. Although hygrometric performance of combined HMEs is slightly lower than that of heated humidifiers, they are clinically as efficient and do not cause more endotracheal tube occlusions. They may be used in the vast majority of medical (including COPD) and surgical ICU patients and for the entire duration of mechanical ventilation. Specific situations, however, such as menacing respiratory acidosis during mechanical ventilation of patients with acute asthma or very severe acute respiratory distress syndrome, require the use of a heated humidifier until the respiratory condition improves sufficiently to enable the use of an HME. Use of heated humidifiers does not increase the risk of ventilator-associated pneumonia, nor does prolonged use of HMEs. However, because circuits get rapidly contaminated with heated humidifiers (unlike HMEs), they carry the potential for cross contamination. Finally, costs of mechanical ventilation are greatly reduced with the use of HMEs.

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